IMPULSE V1: Inferred Microdebris Population from Unmaneuvered LEO State Evolution

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The orbital debris environment below the cataloging threshold remains poorly constrained. NASA's Orbital Debris Engineering Model (ORDEM) and ESA's Meteoroid and Space Debris Terrestrial Environment Reference (MASTER) predict substantially different microdebris fluxes in critical size and altitude regimes, yet independent observational constraints are scarce. Dedicated debris sensors are expensive and spatially limited, while conventional space surveillance cannot resolve individual objects below approximately 10 cm.

IMPULSE proposes an alternative: using decommissioned, non-maneuvering satellites in Low Earth Orbit as opportunistic, distributed sensors. The core idea is that long-term Two-Line Element (TLE) orbital histories contain measurable signatures of collisions or close interactions with untracked microdebris, provided a sufficiently accurate null dynamical model can isolate unexplained impulses from known environmental forces. By reconstructing each satellite's expected orbital evolution, identifying statistically significant deviations, and aggregating across a fleet, IMPULSE attempts to infer the underlying event rate and compare it against model predictions.

This paper presents the completed calibration, validation, and fleet-scale application of IMPULSE V1. We construct a 9-state augmented orbital filter in non-singular equinoctial elements, implement a Fraser–Potter fixed-interval smoother with first-order Gauss–Markov dynamic model compensation, and validate the detection pipeline through a four-stage ladder: null tests, bias recovery, single impulse injection and recovery, and full clean-versus-debris detection demonstrations. We characterize the detection efficiency surface via synthetic Δv injection across a stratified fleet sample, quantify the false-positive budget from drag uncertainty, solar radiation pressure model error, and TLE noise, and propagate all uncertainties through to the primary estimand: the debris-induced event rate λimpulse.

The pipeline recovers synthetic impulses of Δv50 = 0.041 m/s (synthetic clean data) and Δv50 = 0.0385 m/s (real TLE data) with 99.97% confidence, while the aggregate false-positive rate is bounded at 19.46% per window from scrambled TLE noise and approximately 15% per window from in-window baseline triggers on real TLEs. We establish a pre-specified Bayesian decision framework for the ORDEM/MASTER comparison and demonstrate, via synthetic-universe injection, that the pipeline possesses the statistical power to discriminate the models under idealized conditions. However, the fleet-scale application reveals a fundamental limitation: the observed trigger rate of 51.55% dwarfs both the ORDEM-predicted debris signal (5.36%) and the MASTER-predicted signal (0.04%), with a net excess of 32.09% per window attributable to non-debris systematics. The end-to-end uncertainty propagation yields a total measurement uncertainty that is dominated by systematic confounders rather than Poisson counting statistics.

The final result is scientifically meaningful: IMPULSE V1 demonstrates that passive orbital histories contain detectable structure, but that TLE-based dynamical anomaly detection under the baseline operating regime is fundamentally limited by an irreducible systematic floor. We quantify this floor, identify its sources, and chart a path toward V2 strategies that may break the degeneracy

Full technical report: IMPULSE V1: Quantifying Systematic Limits of TLE-Based Passive Orbital Sensing for Microdebris Environment Model Discrimination. https://doi.org/10.5281/zenodo.23852318